



Tamping Assistant (tmA²)

Installation Guide

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Reviewed by: Michael Söllinger

Released and Approved by: Michael Berghuber

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Revision History

Version	Date	Author	Changes made
1.0	April 20, 2023	Mohammad Shahid / Tanmay Mallick	Creation

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1 About tmc

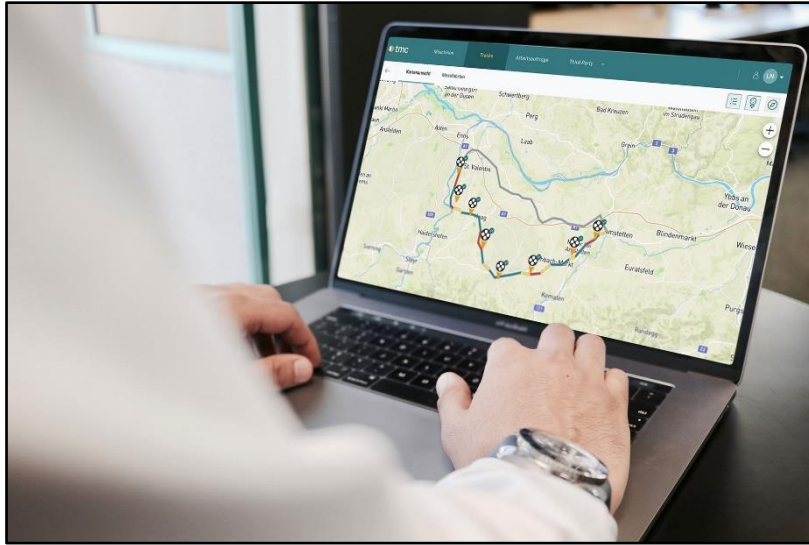


Figure 1. Digital track maintenance - System solutions for maximum efficiency and availability in rail-infrastructure maintenance

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Founded in Austria, headquartered in Linz, and efficiently backed by its subsidiary Vienna and Delhi offices, tmc employs over 100 of the world's brightest scientists, developers, project managers and railway digitalization enthusiasts. An established technological force, tmc covers customer needs across Europe, North America, Japan, and Australia.

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2 About this Manual

Title: Tamping Assistant (tmA²) Installation Guide

Category: Machine

Product: tmA² App

Applicable product release version: 2.0

Required release date: 20.04.2023

Owner: Michael Berghuber

2.1 Intended Purpose

The purpose of this manual is to provide all the necessary information to install the tmA² Application and to integrate its hardware equipment. This manual assumes that the reader has adequate knowledge of the tamping process, tamping machines, and their working units.

2.2 Target Audience

The target audiences for this manual are:

- Service Technicians
- Electrical Engineers

2.3 Reference Information

Refer **tmA² - User Manual** for the usage and operation of tmA² system.

2.4 Point of Contact

The organization that requires coordination between the project and its specific support function:

Headquarter Linz:

Regensburger Str. 3
4020 Linz

Office Vienna:

Am Tabor 42
1020 Vienna

Office India:

801 & 802, 8th Floor, South Wing, Tower 2,
L&T Business Park, Faridabad, Haryana 121003

Phone: +43 732 931659

Email: office@tmconnected.com

2.5 Document Conventions

In this document, certain words or phrases can be represented in different fonts or styles. This highlighting is systematic, with following predefined conventions:

Convention	Description
<i>Italic font style</i>	Italicized text signalizes <i>newly introduced terms</i> or <i>gently emphasized words</i> .
Bold font style	This style is for strongly emphasized words .
<u>bold underline font style</u>	Use this style for referencing to other sections, tables or figures. References also have to be linked to the corresponding source. <i>Example:</i> For more information, see <u>Figure 1</u> .
Courier New	Text written in <code>Courier New</code> signalizes file paths, commands, or an expected output from a console. <i>Example:</i> The file can be found in <code>C:/Windows/recordings/</code>
[KEY]	This style signalizes the action to press a [KEY], a [SEQUENCE] -> [OF] -> [KEYS] or a [KEY] - [COMBINATIONS]. <i>Example:</i> Press [ALT] - [A] to activate an overlay
<Replaceable>	All sections in this style are meant to be replaced by the user. Can also be in <code><Courier New></code> when part of, e.g. a command.
DEVICE NAME	This style is defined for device names represented in ALL CAPS. <i>Example:</i> MDC.
UI Terms	This style is for all User Interface terms represented as <Initial Caps + Bold >. Link multiple steps of navigation with >. <i>Examples:</i> RES1 , SIM CARD .

2.5.1 Box-shaped Highlights

Additionally, this document includes box-shaped areas, highlighting relevant information to the reader with several levels of importance:



Note: This box signalises important information, e.g., memory limitations, user scenarios...



Tip: Here you can find additional information such as alternate navigation, shortcuts, and techniques...



Caution: This one is signaling a precaution to be taken to avoid minor/moderate injury or something that can be problematic for a smooth operation of the application...



Warning: Highlighting an important action to be avoided, which could potentially result in physical harm to user or hardware or can inflict serious issues to the application or its intended workflow, like loss of data or long downtimes of the operability...

3 Preinstallation and Deployment Requirements for the tmA²

This section describes the preinstallation and deployment requirements for tmA² system and hardware equipment.

3.1 Understanding the tmA²

The tmA² system provides support for tamping in turnouts and crossings. It controls the working units (for example, Tamping Unit, Lifting-Lining Unit, Additional Lifting Unit, Dynamic Track Stabilizer Unit, Side Compactor Unit, and Sound Insulator Unit) of a tamping machine during turnout tamping. The system recommends specific actions that must be confirmed by you before they are carried out. The suggestions reduce the risk of operating mistakes significantly, but you can intervene manually at any time.



Figure 2. tmA² Application View on the Display

The system scans the turnout and evaluates the data. It recognizes track parts, rail sides, switches, and obstacles and recommends corresponding actions for the working units. The recommendations are displayed on the tmA² monitor.

The system supports the automatic stop of the machine above the sleepers (tamping positions) and the automatic positioning of all supported working units. While the system is active, you can intervene manually at any time.



Note: All photographs of the tamping machine and machine parts are symbolic images and may be different from the actual machine configuration.

3.1.1 Setting up the Product Environment

The subsequent sections offer an outline of the necessary hardware components for the tmA² system. The tmA² system enables a machine-to-machine (M2M) software interface by supplying the following:

- Recorded sensor data that has been reduced and pre-processed.
- Actions recommended by the tamping assistant.
- Actions executed by the operator.

The tmA² system consists of the following components:

1. **LiDAR/360 Laser Scanner**
2. **Line Scanners**
3. **tmA² Computer**
4. **RGB Camera**
5. **Touch Display Screen**
6. **Additional Hardware**
7. **Optional Hardware**
8. **Tamping Protocol Module (tmOS based)**

3.2 System Requirements

The figure below shows how the hardware components of the tmA² system are configured.

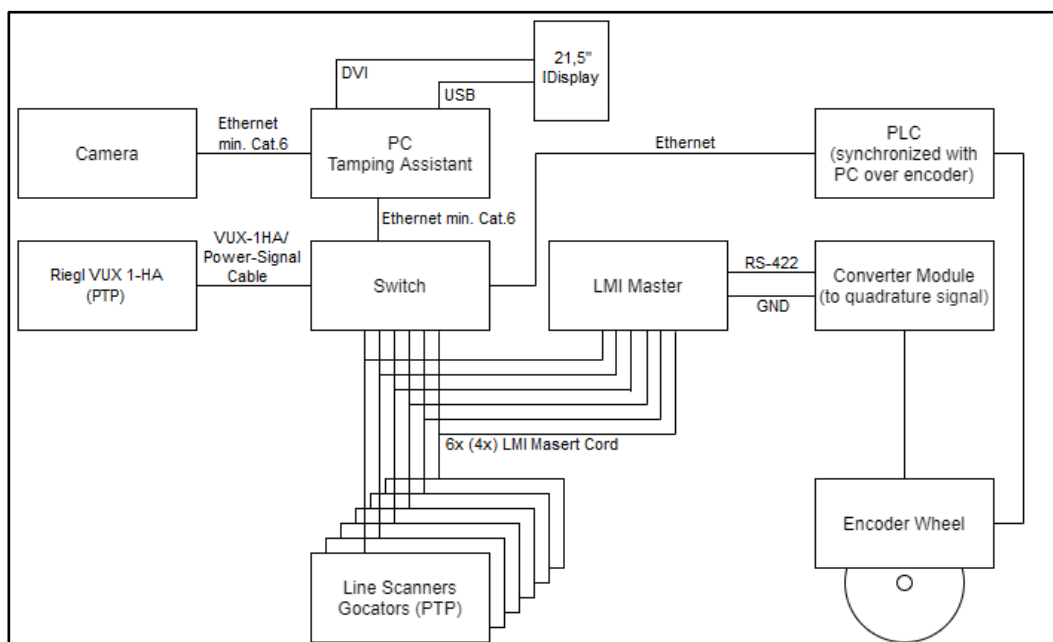


Figure 3. tmA² System Hardware Setup

3.2.1 Hardware Requirements

The following sections provide an overview of the hardware which is required for the tmA² system.

3.2.1.1 LiDAR/360 Laser Scanner

The Riegl VUX-1HA LiDAR Scanner is equipped with PTP capable firmware and can be outfitted with the Protective Housing (PH-VUX) for enhanced durability in harsh environmental conditions.



Figure 4. LiDAR/360 Laser Scanner

3.2.1.2 Line Scanners

For track tamping machines, two units of LMI Gocator with firmware higher than v5.0 are required for each rail, whereas for switch tamping machines, three units of LMI Gocator with firmware higher than v5.0 are necessary per each rail. In addition, LMI Gocator Master 810 (Sensor Switch) is also available.



Figure 5. Line Scanner Gocators

3.2.1.3 tmA² System - Computer

Table 1 shows the minimum hardware requirements for the tmA² computer and how some components should be adjusted.

Table 1. tmA² System - Computer Requirements

Component	Details
Operating System	Window 10 (Pro or IoT 64 bit)
CPU	8 Cores / 3.7 GHz
RAM	64 GB
Memory	System SSD 1 TB  Tip: Use a replaceable 4 TB SSD for recording data storage and another replaceable 4 TB SSD for transferring data (strongly suggested).
Network Card	Ports 3GigE / RJ45

GPU	 Tip: A basic Tamping Protocol can be operated without an assistant function and doesn't require an additional graphic card. However, it is recommended to have one for obstacle detection and visualization features in the protocol.
-----	--

3.2.1.4 RGB Camera

The RGB camera being used is the FLIR ORX-10G-89S6C-C, which has a resolution of 8.9 megapixels and a frame rate of 93 frames per second and is mounted using a C-Mount. It includes a lens, cable, host adapter, power supply, and protective housing, and is powered according to the manufacturer's specifications with proper illumination. It is used for recording purposes.



Figure 6. RGB Camera

3.2.1.5 Touch Display Screen

The touch display screen features a minimum 21-inch screen size with Full HD 1920 x 1080 resolution and must be mounted in portrait orientation, if possible, with the necessary USB connectivity to enable touch support.

3.2.1.6 Additional Hardware

The system requires the use of various additional hardware components, including LMI Master Cord set, RIEGL Power-Signal-Cable, Ethernet CAT 6 Cable, measuring wheel, TTL RS422 Converter Modul, Gig-Ethernet Switch, and optional USB Cable (tmA² computer to tmA² touch panel) to ensure proper connectivity and functionality.

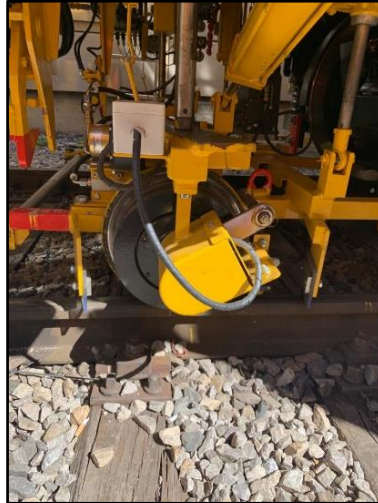


Figure 7. Measuring Wheel with Encoder

3.2.1.7 Optional Hardware

The current system includes an optional hardware component, the Plasser Tie Finder with LAN interface, which allows for efficient and accurate detection and identification of railway ties.

The **Tie Finder** is a controller that has microcontroller module with 8 digital and 8 analog I/O. It processes the analog signals from 2 threshold sensors, which are attached to the left and right of the front measuring carriage and detect the metal attachments.



Figure 8. Tie Finder

3.2.2 Software Requirements (Optional)

A detailed overview of the software required by the tmA² system is provided.

tmOS Web-App (Tamping Protocol Module)

- Docker
- Kafka.

Refer to [tmOS Installation Guide](#) to configure tmA² (Tamping Protocol) on tmOS Web-App.

4 Installing the tmA² Application

This section provides detailed information about how to install tmA² application on windows operating system.

4.1 Installing tmA²

Follow the below steps to install the Tamping Assistant (tmA²) application:

1. Shut down any processor on the machine.
2. Double-click **TampingAssistant x.x.exe**.
The **Windows Defender** window is open on the display.

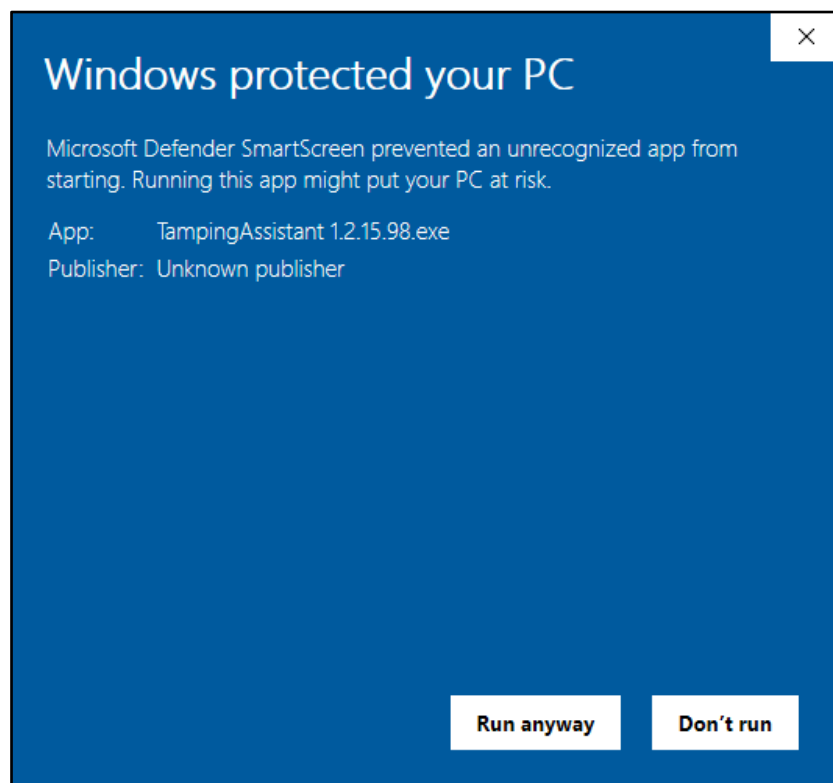


Figure 9. Windows Defender Window

3. Click **"Run Anyway"**.
4. Click **Browse** to choose or change the application destination folder. By default, the application folder is stored in C drive.
Click **Next** to go through each window in turn.

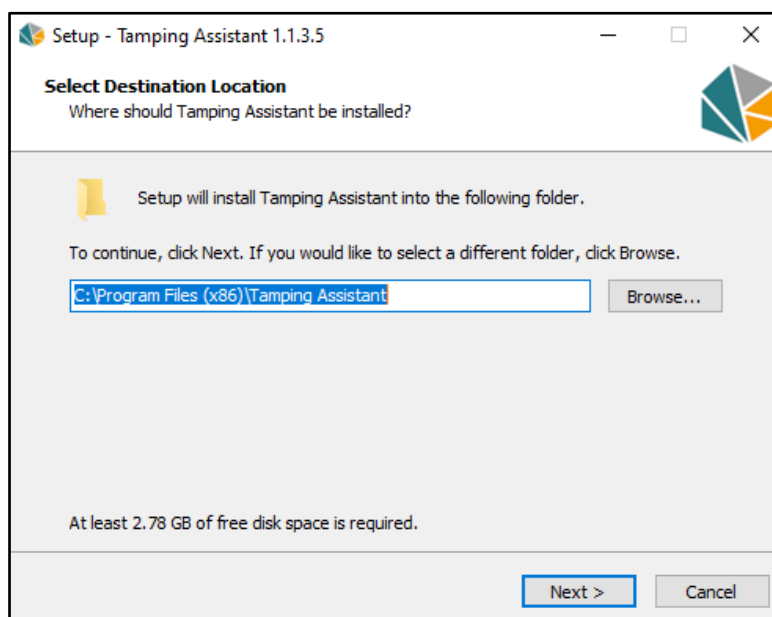


Figure 10. Select Destination Location

5. Select **Full installation** component and click **Next** to continue.

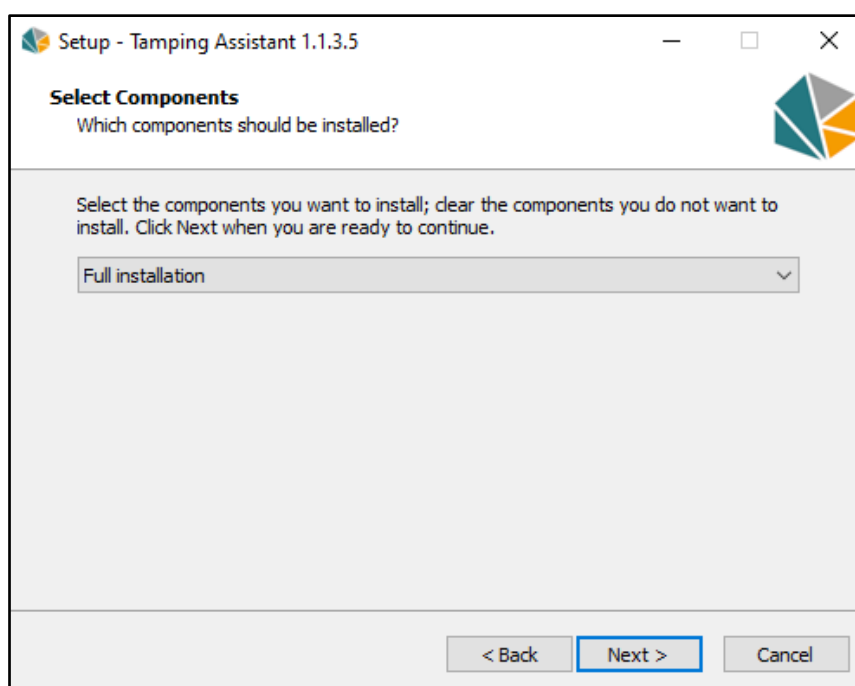


Figure 11. Select Components (Full Installation)



Note: If the application is already installed on your system, refer to **Upgrading from Previous Versions** section to upgrade the existing version of tmA².

6. Select the additional check boxes and click **Next** to continue.

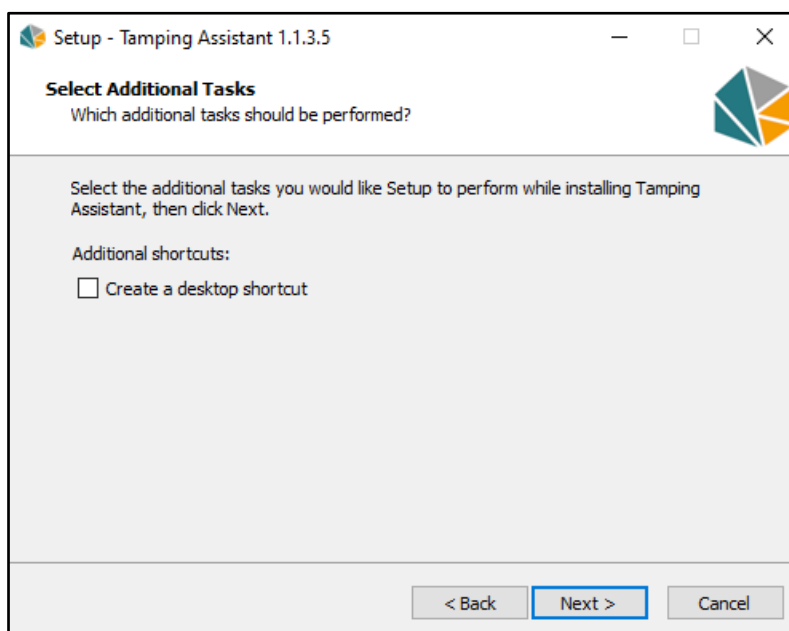


Figure 12. Select Additional Tasks



Note: Auto Start the tmA² application when Windows starts, and user logged in.

7. Click **Install**.

The setup is installing on your computer.

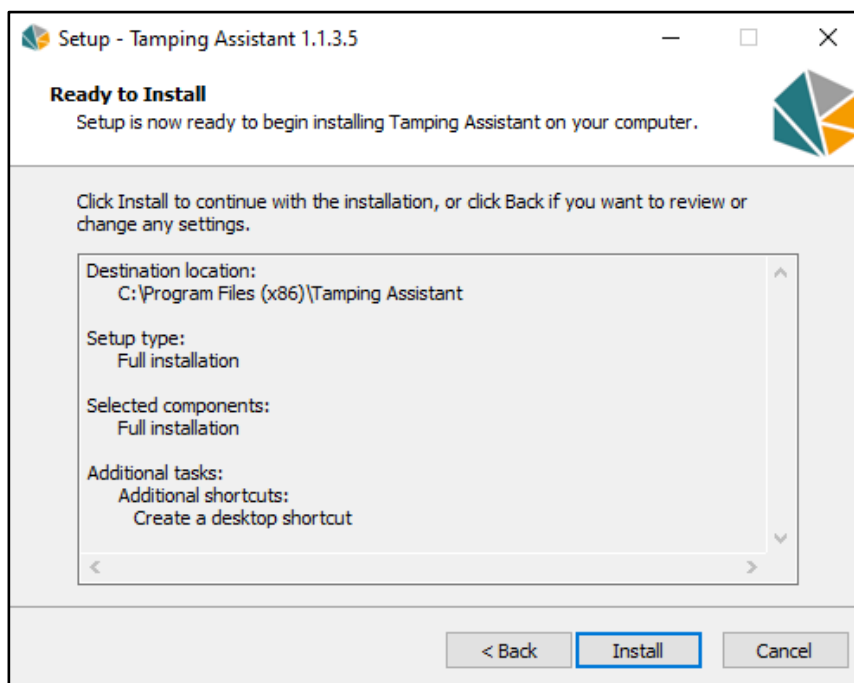


Figure 13. Ready to Install



Caution: Once the installation is in progress, do not click the **Cancel** button. This may lead you to stop the installation progress and need to start from the beginning.

8. Wait for the installation to complete.

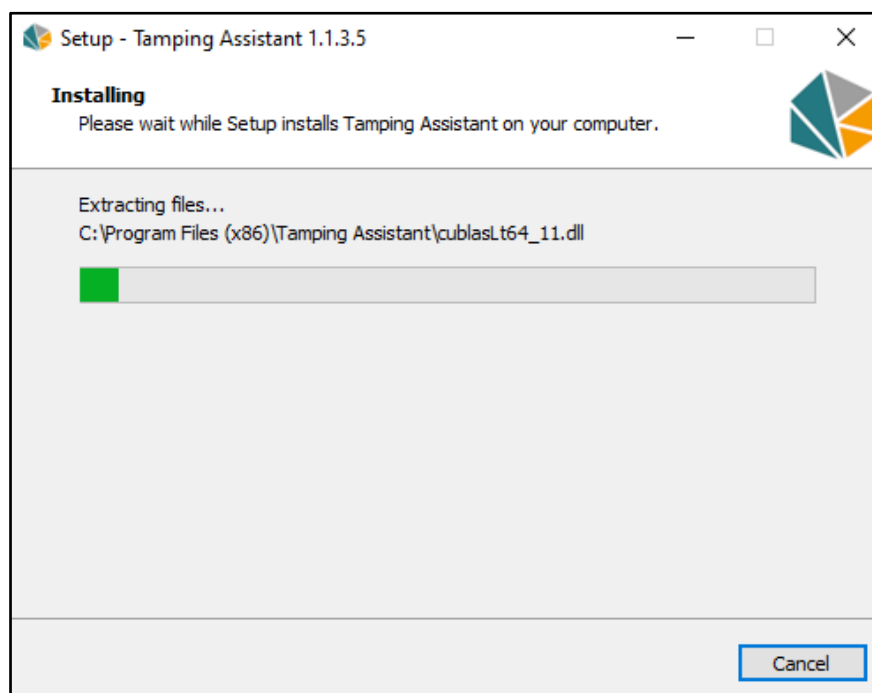


Figure 14. Installing the Application

9. **Figure 15** confirms that the installation has been completed.

Click **Finish** to complete the installation procedure.



Figure 15. Installation is Completed

4.2 Configuring tmA² Hardware



Note: Please refer the **Configuration & Calibration LIDAR Scanner (Tamping Assistant) & Configuration & Calibration Lines Scanners (Tamping Assistant)** for more details about configuration and calibration.

The tmA² hardware are assigned with IP addresses which are required to be configured as per the below table.

Table 2. Hardware IP Address

Name	Sensor IP	Adapter IP - tmA ² PC	Subnet Mask
Gocator left rail additional	192.168.40.10	192.168.40.1	255.255.255.0
Gocator left rail left	192.168.41.10	192.168.41.1	255.255.255.0
Gocator left rail right	192.168.42.10	192.168.42.1	255.255.255.0
Gocator right rail left	192.168.43.10	192.168.43.1	255.255.255.0
Gocator right rail right	192.168.44.10	192.168.44.1	255.255.255.0
Gocator right rail additional	192.168.45.10	192.168.45.1	255.255.255.0
LIDAR Scanner	192.168.233.111	192.168.233.113	255.255.255.0
LIDAR Scanner Housing	192.168.233.112		255.255.255.0
RGB Camera	DHCP	DHCP	255.255.255.0
PLC	192.168.0.1	192.168.0.80	255.255.255.0
tmOS - PC	192.168.0.81	192.168.0.80	255.255.255.0
ALC - PC	192.168.0.202	192.168.0.80	255.255.255.0

4.2.1 Configuration and Calibration of LiDAR Scanner

To configure the LiDAR Scanner and Housing, it is required to connect the LiDAR scanner to LMI Master Hub.

4.2.1.1 Setting IP Address of tmA² PC

The factory IP address of a new LiDAR Scanner is 192.168.0.128 or if it is in the housing 192.168.0.234. The tmA² PC needs to set the adapter setting of the LMI master hub (sensor switch), to reach that IP address.

1. Login to tmA² PC.
2. Select **Control Panel > Network and Internet > Network and Sharing > Change Adapter Settings**.
3. Select Sensor Switch **Ethernet Port > Properties**.
4. Internet protocol **IPv4 > Properties > Advanced**.
5. Make sure that all the IP address of all Line scanners and the LiDAR scanner are listed.
6. Add a new IP address with the following inputs:
 - o IP address: 192.168.0.111
 - o Subnet mask: 255.255.252.0

4.2.1.2 Setting IP Address of LiDAR Scanner

1. Connect the LiDAR Scanner to the LMI Master Hub (Sensor Switch).
2. Ping 192.168.0.234 in command line to establish the connection with LiDAR Scanner.
3. Start **Scanner Application > Tools > Configure Riegl Vux**.
4. Settings:
 - Current IP : 192.168.0.234
 - Target IP : 192.168.233.211
 - Subnet : 255.255.252.0

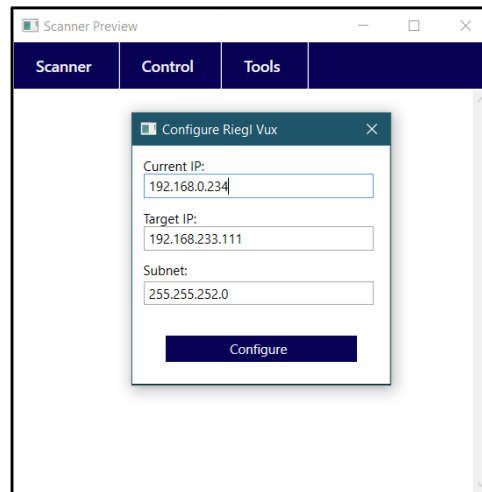


Figure 16. LiDAR Scanner IP Configuration

4.2.1.3 Calibration of LiDAR Scanner



Tip:

For more information on Calibration of LiDAR Scanner, see the latest **Configuration & Calibration LIDAR Scanner (Tamping Assistant)** Guide.

4.2.2 Configuration of RGB Camera

The RGB camera is set to auto-exposure mode. To connect the RGB camera with the tmA² application, you must have the Vimba driver installed. Follow these steps in the Vimba viewer:

1. Set exposure-mode to **Auto-Exposure**.
2. Set User-Profile to a non-default-profile.
3. Set Exposure-Auto to Continuous.
4. Save User-Settings.

4.2.3 Configuration and Calibration of Line Scanner(s) (Gocators)

The following section contains detailed information on how to configure and calibrate the Line Scanner(s).

4.2.3.1 Setting IP Address of tmA² PC

The factory IP address of a Line Scanner is 192.168.1.10. The tmA² PC needs to set the adapter setting of the LMI master hub (sensor switch), to reach that IP address.

1. Login to tmA² PC.
2. Select **Control Panel > Network and Internet > Network and Sharing > Change Adapter Settings**.
3. Select Sensor Switch **Ethernet Port > Properties**.
4. Internet protocol **IPv4 > Properties > Advanced**.
5. Make sure that all the IP address of all Line scanners and the LiDAR scanner are listed.
6. Add a new IP address with the following inputs:
 - IP address: 192.168.1.5
 - Subnet mask: 255.255.255.0

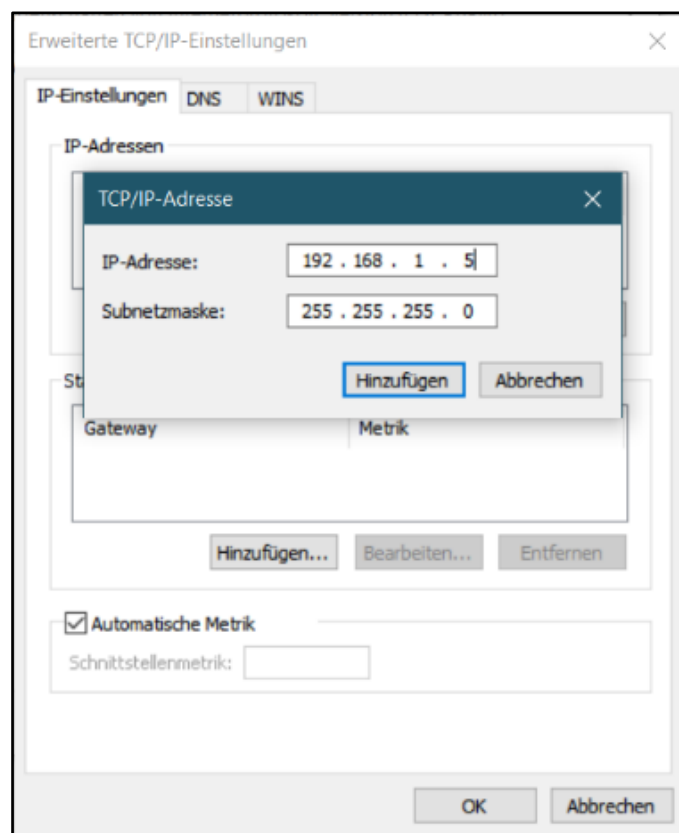


Figure 17. Setting IP Address of Line Scanner

4.2.3.2 Setting IP Address of Line Scanner



Note: The Line Scanner cables are colored black at the connection with Network switch and gray at the connection with LMI Master Hub.

1. Disconnect all line scanner cables from the LMI Master Hub (Sensor-Switch) to begin IP configuration.
2. Connect only one line scanner cable at a time to configure its IP address.
3. Ensure that the cable being connected is not the machine-network or LiDAR Scanner cable.

4. Use the command line to ping the IP address 192.168.1.10 and establish a connection with the line scanner.
5. Access the web interface at 192.168.1.10 via web-browser to check the description of the connected line scanner.
6. Activate the Machine Laser Safety Switch to ensure safe operation.
7. Set the Trigger-Mode to **Time**, otherwise the line scanner will not produce any images.
8. Use the LMI-Discovery tool to set the IP address according to the IP-Address Concept.
9. Disconnect the current cable and repeat the configuration steps for all Gocators.
10. Once all devices are configured with their respective IP addresses, as per the IP-Address Concept, they can be connected to the switch.

4.2.3.3 Sensor Settings

You can access every Gocator device through its IP address using a web browser. If you haven't activated any Gocators yet, you can find the factory IP address and use it to access the device.



Tip:

For more information on Sensor Settings of LiDAR Scanner, see **Section 6** of the latest **Configuration & Calibration Line Scanners (Tamping Assistant)** Guide.

4.2.3.4 Calibration of Line Scanners



Tip:

For more information on Calibration of Line Scanner, see **Section 7** of the latest **Configuration & Calibration Line Scanners (Tamping Assistant)** Guide.

5 Upgrading from Previous Versions

**Tip:**

Before starting the upgrade, it is recommended to stop the tmA² Windows Service then the installer must be run again.

If the installer detects a previous version of the application is already installed on machine, it prompts to select **Full installation** and **Update installation** components. If **Update installation** is selected, it redirects you to upgrade the application. All other application settings and machine parameters remain the same.

5.1 Upgrading tmA²

Follow the below steps to upgrade the Tamping Assistant (tmA²) application:

1. Shut down any processor on the machine.
2. Double-click on the **TampingAssistant x.x.exe**.
3. If there is any previous version of application exists, the screen shown in **Figure 18**. Update installation will appear.

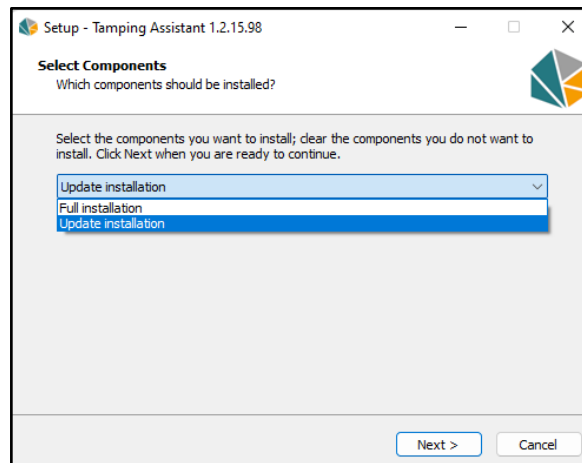


Figure 18. Update installation

4. Select **Update installation** component and click **Next** to continue.
5. If the **Components Exist** dialog box is displayed as shown in **Figure 19** and click **Yes**.

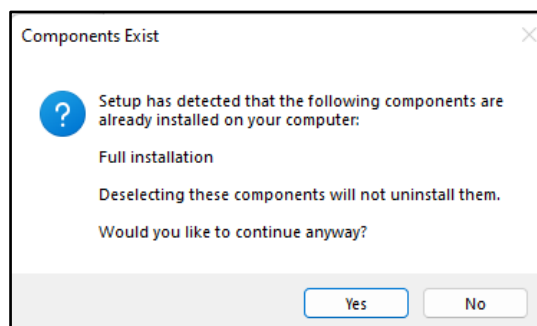


Figure 19. Update installation - Dialog Box

6. Select the additional check boxes and click **Next** to continue.

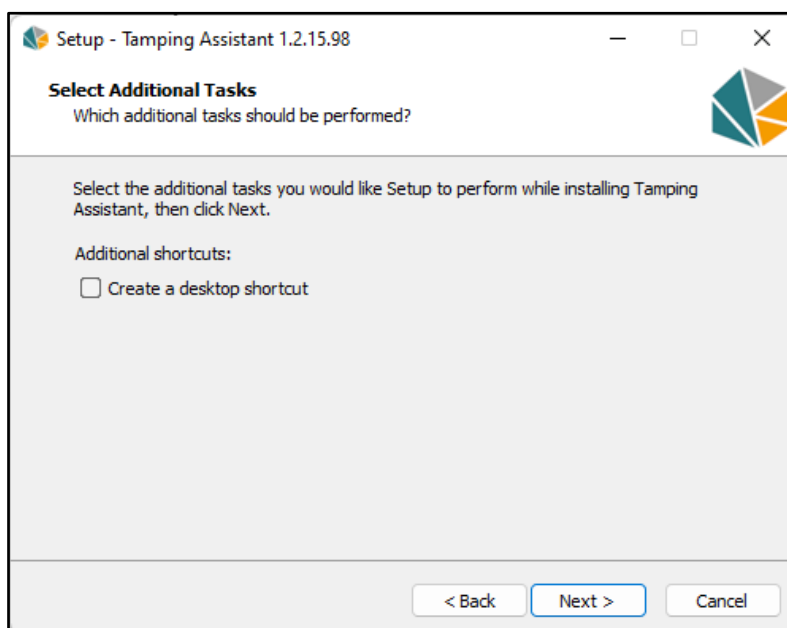


Figure 20. Select Additional Tasks



Note: Auto Start the tmA² application when Windows starts, and user logged in.

7. Click **Install**.

The setup is installing on your computer.

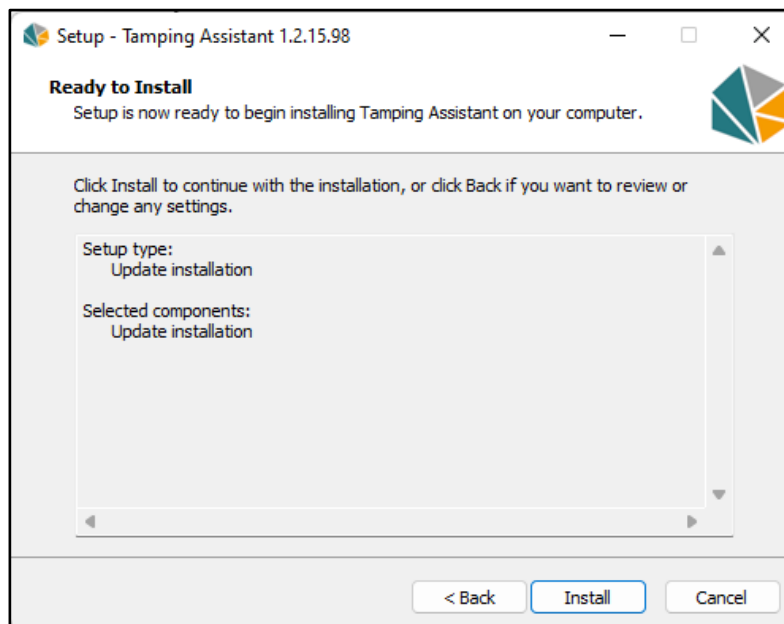


Figure 21. Ready to Install



Caution: Once the installation is in progress, do not click the **Cancel** button. This may lead you to stop the installation progress and need to start from the beginning.

8. Wait for the installation to complete.

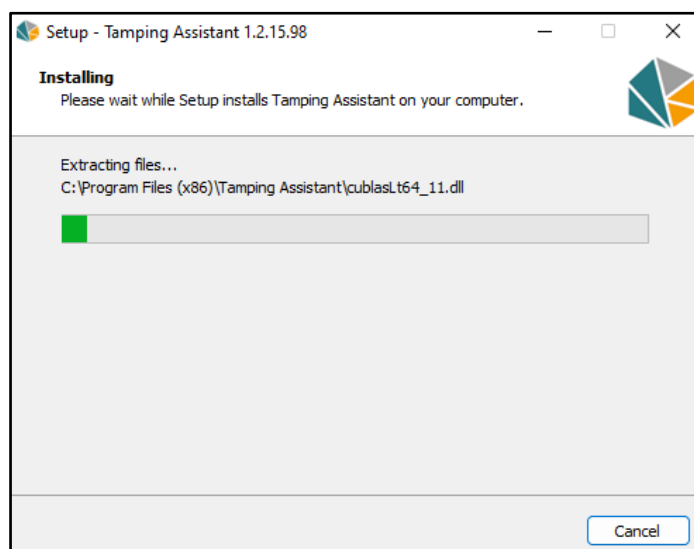


Figure 22. Installing the Application

9. The setup display confirms that the installation has been successfully completed (Refer **Figure 23**).
10. Click **Finish** to complete the upgradation procedure.

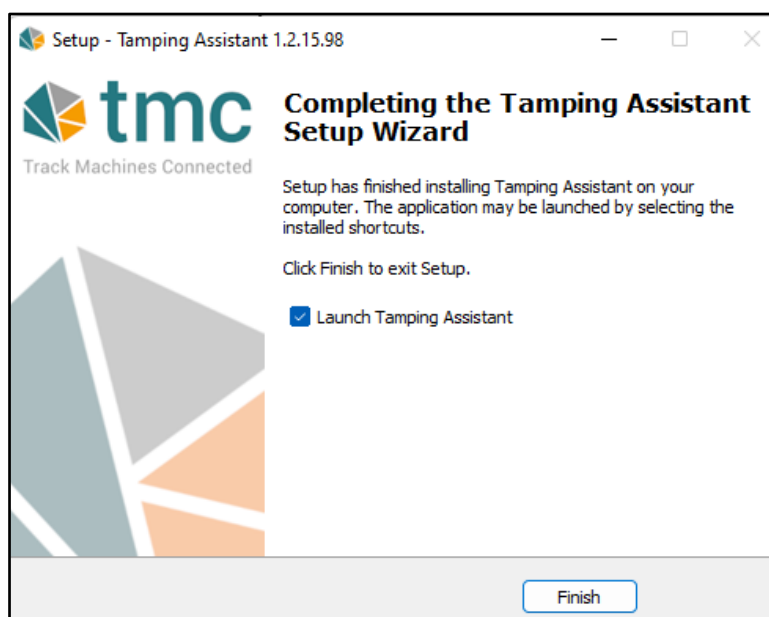


Figure 23. Installation is Completed

6 Uninstalling the tmA² Application

This section describes how to remove the tmA² application from the computer and delete the Tamping Assistant folder in the C drive.

To uninstall the application,

1. Double-click Control Panel > Programs and Features > DRP > Uninstall.

The **Uninstall** dialog is open on the display.

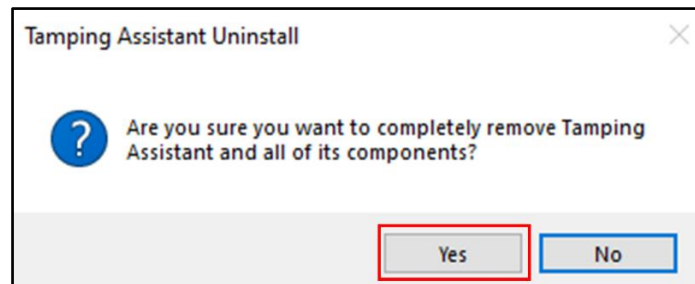


Figure 24. Uninstallation Confirmation

2. Click **Yes**.

This permanently removes the installer as well the folder and you will get a confirmation message on the display.



Note:

This uninstallation removes only the Tamping Assistant application, not the entire file structure. The measurement files are kept and deleted manually. Settings and any other files are deleted.

7 Troubleshooting

7.1 Maintenance Information

For safe usage of the tmA² system, you must adhere to the following instructions during installation:

1. Ensure that the measurement device is used in a clean environment, free from dust generated during tamping.
2. Avoid exposing the Gocator measuring areas to direct sunlight.
3. Operate within the temperature range specified in the Sensor Specs.

7.2 Frequently Asked Questions

1. What is the use of tmA² Application?

Answer: The tmA² application provides machine operators with step-by-step guidance during the handling of machines and provides support for tamping in straight tracks, turnouts, and crossings.

2. What is the process to obtain the installation file for the tmA² application?

Answer: To obtain the installation file for the tmA² application, you need to contact the tmc support team. They will provide you with the necessary installer file, which you can then download and install on your computer.

3. Do the files need to be downloaded to a particular directory?

Answer: The application can be downloaded and installed from any directory from your computer.

4. Do I need administrative privileges to install the tmA² application?

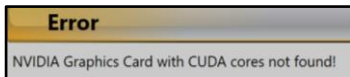
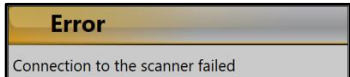
Answer: You may require administrative privileges to install the application. It is recommended to check the support team for specific guidance.

5. What should I do if I encounter issues after installation?

Answer: If you encounter issues after installation, you can try restarting your computer, checking for updates, or contacting the support team for assistance. Providing detailed information about the issue, such as error messages or screenshots, can help the support team diagnose and resolve the issue more effectively.

7.3 Troubleshoot Instances with Failed Status Checks

The following table provides details of various failure status displayed:

Error Status	Issue Description	Workaround / Solution
	During the first run of the tmA ² application, user may face below errors.	In case, if you get this error message then you have to install NVIDIA Graphics Card. Check the status of Windows driver and re-install the driver. Check for the defective GPU.
	Scanner Connection Failed	After installation of Line Scanner, it should be configured.

Error PLC connection failed	PLC Connection Failed	Check for any hardware connection issues with PLC.
Error tmOS connection failed to establish	tmOS Connection Failed	Do settings for the tmOS Kafka interface with Tamping protocol.

7.4 Technical Support and Service

For product support and service, refer to the link below:

<https://tmconnected.com/support>

8 Glossary

Term	Definition
Additional Lifting Unit	Additional Lifting Unit lifts the third rail in the area of the long sleepers at the turnouts. Moreover, the lifting process synchronizes with the Lifting-Lining Unit of the tamping machine.
Graphics Processing Unit (GPU)	The graphics card is responsible for rendering an image to your monitor. There are two types of graphics cards - Integrated and Discrete.
Lifting-Lining Unit	Lifting-Lining Unit helps to lift and hold the track in the corrected position while being tamped.
Programmable Logic Controller (PLC)	Executes actions suggested by tmA ² and provides feedback based on performance and changes.
Solid State Drive (SSD)	It is a new generation storage device used in computers. Generally, it speeds up computers due to low read-access times and fast throughputs.
Tamping Mode	The mode set on the Touch Panel, for example, main track, diverging track, and the turnout direction.
Tamping Unit	Tamping Unit supports tamping various types of sleepers. For example, single and double sleepers.
Working Units	Units support tamping work, for example, Tamping Unit, Lifting-Lining Unit, Additional Lifting Unit.

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9.3 Referenced Documents

Name	Description
tmA ² - User Manual	User Manual for tmA ² system
Track Machines Operating System (tmOS) User Manual	Deployment of tmOS and System requirements
Configuration & Calibration Guide - LIDAR Scanner	Configuration & Calibration details of LIDAR Scanner
Configuration & Calibration Guide - Line Scanner	Configuration & Calibration details of Line Scanner

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